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Remote Sensing C
Test Packet

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Instructions

1. You will have 50 minutes to take this test.
2. Answer all questions on your Answer Sheet. The space provided for each question is sufficient to fit a response earning full points.
3. All questions have their value indicated in parentheses, both in this Test Packet and on the Answer Sheet. This test has a total of 200 points.
4. You are free to write on and take apart the pages of the Test Packet or Image Sheet as you wish.
5. For numerical questions, answers within a reasonable range will earn full credit - accounting for significant figures is not required.
6. Ties will be broken by comparing points earned for each question in order.

Section A - 10 points

Suppose you are trying to measure N_2O in the atmosphere. **Image 1** shows some data relating to N_2O and the general atmosphere.

1. What is the historical chemical name of N_2O? (1)
2. What region of the electromagnetic spectrum is shown in Image 1? (1)
3. What type of graph is shown in Image 1? (2)
4. What happens to light at the orange peak at approximately 4.4 microns in relation to N_2O? (3)
5. Consider the total atmosphere data (purple) in Image 1. The dominant cause of atmospheric absorption is a result of which substance? (2)
6. The absorption in this part of the electromagnetic spectrum is primarily a result of what molecular effects in the substance from the previous question? (1)

Section B - 10 points

Consider a satellite flying eastward in an equatorial orbit, which is using Doppler radar to determine its speed relative to the surface.

7. Suppose the satellite sends out radar at a frequency of 5 GHz and receives a return signal of 5.0005 GHz. What is the ratio between the return and outgoing frequencies? (1)
8. What is the speed difference between the satellite and the Earth's surface? Disregard the angular component (altitude effects) and curvature of the Earth for your calculations. (4)
9. The Earth has a radius of approximately 6,378 km at the equator. What is the rotational speed of the Earth's surface at the equator? (3)
10. What is the orbital speed of this satellite? (2)

Section C - 20 points

Chlorophyll concentration is a particularly important indicator of the Earth's current global productivity. Let's talk about how it's measured.

11. Examine **Image 2**, the spectral signature of chlorophyll-a. Approximately at what two wavelengths does it absorb blue and red light, respectively? (2)
12. **Image 3** shows spectra from four samples of seawater with varying concentrations of chlorophyll. Please explain, in terms of radiance differences, why water with high chlorophyll concentration often appears as a shade of green rather than blue. (3)
13. **Image 4** presents data collected by PACE displaying chlorophyll concentrations within the ocean over a window in late 2024. In essence, chlorophyll concentrations are another way of measuring the concentration of what? (1)
14. What is one advantage of PACE's Sun-synchronous orbit? (1)
15. Describe why the areas surrounding the borders of continents have extremely high chlorophyll concentrations. (2)
16. This image does not display data towards the north pole; what would data likely look like in the areas currently not filled in the northern hemisphere? Why can data not be displayed for the entire globe at once? (3)
17. Commonly, collecting chlorophyll-a concentrations involves data from the penetration depth, or z_{90}. Define the penetration depth, or z_{90}. (2)
18. Some algorithms depend only on surface chlorophyll-a concentration. In an area with a well-defined subsurface Chl-a maxima layer (SCML), explain why this measurement would be inaccurate. (3)
19. Explain how the location and structure of the SCML can be accurately modeled. (3)

Section D - 20 points

Refer to **Image 5** for these questions. This image shows data collected by CERES, averaged over many years.

20. Does this image show shortwave or longwave radiation? (1)

21. What is the origin of the radiation from Image 5? (2)

22. What property of objects determines the value of the measurement in Image 5? (2)

23. Consider the light-colored area near the equator across most of the image, e.g. near the letter A. What climate phenomenon causes this area to be light-colored? (2)

24. How does the Earth's atmospheric circulation produce the phenomenon from the previous question? (2)

Consider also **Image 6** for these questions. This image also shows data collected by CERES, averaged over many years.

25. Does this image show shortwave or longwave radiation? (1)

26. Do the yellow areas on this map indicate high or low radiation? (1)

27. The blue/purple regions on this map can be divided into three broad categories corresponding to what surface climatological conditions? (3)

28. What is the reason why each category from the previous question shows up as blue/purple on this map? (6)

Section E - 25 points

Forest fires are an important aspect of moving carbon through earth's systems, and interact closely with earth's atmosphere. Answer the following, using **Image 7** and **Image 8** when referenced.

29. What effect do forest fires have on local atmospheric CO2 levels? (1)

30. OCO-2 has been used to collect atmospheric CO2 data, but that ability is hindered by heightened aerosol concentration. Why is OCO-2s ability to measure atmospheric CO2 hindered by increased aerosol concentration? (2)

31. More properties can be used to model CO2 concentrations as they pertain to forest fires. Reduced reflectance due to smoke cover, in particular, is often used. What instrument uses its eighth band, 405-420 nm, to establish a region of reduced reflectance and therefore demarcating smoke plume area? (1)

32. Examine Image 7. By plotting CO2 concentration against reduced reflectance in the previous question, a relationship between the two is established, which allows for calculation of CO2 concentration regardless of our ability to measure it directly. In order to accurately calculate CO2 emissions from these forest fires, what property must be subtracted from the XCO2 values presented in Image 7? (2)

33. What part of the fire itself is represented by the areas with the highest XCO2 values? (2)

34. Reduced reflectance is also useful for identifying boundaries of previously burned areas. Explain why. (2)

35. Name another property can be observed to demarcate recent burn scars. What would you expect to be changed about this property? What approximate bandwidth, on the same instrument from question 31, is useful for measuring this property? (3)

36. The effects of aerosols on the earth's heat budget are particularly relevant to modeling and studying forest fires. Consider Image 8. The red line represents a scenario where the emission and output of aerosols is closely regulated and limited, while the black line represents a scenario in which there is no aerosol regulation, and greenhouse gases are the main driver of warming. Explain how aerosol mitigation affects surface downwelling shortwave radiation, relative to greenhouse gas warming. (2)

37. What does the evidence presented in Image 8 suggest about the effect aerosol mitigation would have on latent heat flux? (2)

38. Discuss how this aerosol mitigation would affect plant behavior, as well as the forests' susceptibility to fires. (3)

39. What is one way that, at high latitude (> 45 degree), boreal forests, greenhouse gas warming may lead to increased soil moisture? (2)

40. With all this in mind, compare the impact of aerosol concentration and greenhouse gas-driven global warming on forest fire likelihood at higher latitudes. What condition makes forest fires the most likely? (3)

Section F - 25 points

Refer to **Image 9** for these questions. This image shows sea surface temperature data collected by Aqua-MODIS during the night (12 hours) of October 9th, 2025, using reflected light with wavelengths of approximately 4 microns.

41. Was this data collected using active or passive sensing? (1)
42. Consider the measurement scale on the bottom of the chart that indicates a numerical range for the colors. Based on the data shown, what is the most likely unit used to measure this data? (2)
43. Is this data an anomaly measurement? (1)

Consider the black areas on this image marked by the letters A, B and C.

44. What is the reason for the blackness near letter A? (2)
45. What is the reason for the blackness near letter B? (4)
46. What is the reason for the blackness near letter C? (3)

The Earth's circumference at the equator is 40,075 km.

47. What is the swath width of Aqua-MODIS? (3)
48. Based on the data patterns in this image, what type of orbit does this satellite move in? (1)
49. Based on the number of orbits taken by the satellite, what is its orbital period? (2)

Aqua-MODIS measures sea surface temperature in two different regions of the electromagnetic spectrum, at around 4 microns (shortwave IR) and 11 microns (longwave IR) respectively.

50. Consider an emission spectrum at typical sea surface temperatures. What can you say about the positions of the SWIR and LWIR bands relative to the spectral peak wavelength? (2)
51. How does your answer to the previous question indicate the relative sensitivities of these two bands? (4)

Section G - 25 points

The intensity and frequency of El Niño-Southern Oscillation events has changed over recent years; to understand why, let's look at some mechanisms which drive the process.

52. What is the name of the relationship, acting as a primary driving mechanism for ENSO events, that is a positive feedback loop connecting sea surface temperatures, winds, and ocean currents? (2)

53. Stochastic variations in the trade winds cause a weakening in equatorial currents that keep the western Pacific warm; how does this affect the sea surface temperature in the central Pacific? (2)

54. Explain the effect this new temperature structure for the Pacific Ocean has on atmospheric convection and wind activity over the Pacific. (3)

55. Examine **Image 10**. Where the Y-axis is the modelled strength of the aforementioned feedback loop across climactic stages, and "ka" means thousands of years ago, what sorts of climates show the strongest growth rate of sea surface temperature anomalies in the central Pacific? Explain your answer. (3)

56. How is this feedback strength related to the strength of Walker circulation? Consider how this connects to the mixing of shallow to deep ocean water. (2)

57. Using **Image 11**, how is feedback strength related to the frequency of ENSO events considered to be extreme? (2)

58. Discuss why varying climatic states (glacial periods, elevated greenhouse gas concentration) impact the strength of this feedback loop. (4)

59. Examine **Image 12**. These maps show anomalous zonal wind stress onto the zonal gradient of sea surface temperature, which is another way of assessing feedback loop strength. Contours represent data from 1979 to 1999, while shading represents data from 2000 to 2021. How has the position of maximum feedback strength moved over time? (1)

60. The movement of feedback strength results from a change in precipitation anomalies due to adjusting background sea surface temperatures. What change is observed in background temperatures throughout the different regions in the Pacific Ocean? How does that affects precipitation, and why? (4)

61. This interdecadal shift in background temperatures of the tropical Pacific has been thought to explain what two changes in El Niño conditions? (2)

Section H - 30 points

Many considerations need to be made when designing a satellite. Answer the following questions about design choices and specific properties of satellites with respect to measuring a certain quantity.

62. What happens to the ground resolution of an image when the system height of an aerial imaging system is increased? (1)

63. Minimum ground separation is defined as the minimum distance between two objects on the ground at which they can be resolved on the image. Calculate the minimum ground separation for a ground resolution of 10 line-pairs per mm. (3)

64. Describe what might be some of the smaller scale features that could be observed for a satellite with a minimum ground separation like the one you just calculated. (3)

65. As discussed elsewhere in this test, ocean color is a particularly useful metric for studying aquatic ecosystems. Explain why having a higher temporal resolution than that offered by a polar-orbiting satellite for such a measurement might be particularly useful for ensuring data completeness and accuracy. (3)

66. What might one change about this satellite in order to attain this higher temporal resolution? (2)

Lets discuss decisions made in order to observe a particular quantity: precipitation.

67. Originally, thermal IR channels were particularly useful for rainfall estimation. Discuss why this was an effective way making estimations. (3)

68. Is this an example of active or passive sensing? (1)

69. Microwave radiation has also been used to estimate rainfall; why might microwave radiation be more useful for estimation than IR radiation? Explain in terms of the waves themselves. (3)

70. Why is IR radiation still used for measurement? (2)

71. Discuss how the differences in measurement capabilities between IR and microwave radiation affect the orbital conditions of the satellites themselves. (4)

72. What is the maximum latitude which a geostationary satellite would be able to observe? (2)

73. Consider a satellite positioned 800 km above Earth's surface moving in a circular orbit. Calculate a period for this satellite's orbit. Round your answer to the nearest minute. (2)

74. Which wavelength would you expect this satellite to use in order to provide rainfall estimations? (1)

Section I - 35 points

In this section, we will model the Sun and Earth as blackbodies, consider how their interactions affect the Earth's temperature, and add more complex elements to the model to analyze their effects.

75. The Sun's surface temperature is approximately 5800 Kelvin. What is the name of the equation that describes the relationship between the temperature of a blackbody and its radiant flux? (1)

76. What is the radiant flux at the Sun's surface, using the provided temperature? (3)

77. The radius of the Sun is approximately 700,000 km, while the Earth is approximately 150,000,000 km from the Sun. What is the ratio between radiant flux at the Sun's surface, and radiant flux at the Earth's orbital distance? (3)

78. Using your answer from the previous question, what is the radiant flux at the Earth's orbital distance? (2)

79. In order for the Earth's temperature to remain in equilibrium, its outgoing longwave radiation must match the incoming shortwave radiation from the Sun. Using our model, what is the ratio between its shortwave and longwave radiation? (3)

80. Using your answer from the previous question, and the equation from the first question in this section, compute the Earth's temperature in this model. (4)

Now we will begin complicating this model a little further. Suppose that instead of being a blackbody, the Earth has an albedo of 0.3 with respect to shortwave radiation.

81. What does an albedo of 0.3 indicate about the Earth's interactions with shortwave radiation? (2)

82. Does adding this albedo factor to the model increase or decrease Earth's equilibrium temperature? (1)

83. Accounting for this albedo factor, recompute the Earth's equilbrium temperature. (5)

In addition to the Earth's albedo of 0.3, we'll now also consider the impact of a heat-trapping atmosphere. Let's model this atmosphere as a single uniform layer that envelops the Earth at an altitude of zero. Our atmosphere will be transparent to shortwave radiation, but it will absorb 100% of longwave radiation and reemit it equally upward (to space) and downward (back to Earth's surface).

84. Write an equation that expresses the atmospheric flux toward the surface Q_A in terms of the Earth's surface flux Q_E. (2)

85. Write an equation that expresses the Earth's surface flux Q_E in terms of the atmospheric flux Q_A and the solar flux at the Earth's orbit Q_S. (3)

86. Solve the system of equations you wrote in the previous two questions and compute the equilibrium temperature of the Earth. (6)